WP 13 | DISTRICT HEATING SYSTEMS

# Programmable District Heating Thermal Loss and Dispatch Maps with paceval

Two versioned pipe-temperature models, hydraulic pressure proxies and a reproducible two-consumer network screening workflow



Concept illustration; not a computed research result.

## Verified mathematical research outcome

Using the real paceval MCP service, 22 pipe-level mathematical graphs were executed for 49 synthetic edge conditions, and 25 composed network mathematical graphs were executed for 25 dispatch scenarios. The service reported version 4.26 and no per-point errors for these main studies. Independent 90-digit mathematical audits passed for 547 archived output values.

## Scientific scope and limits

The results describe steady-state, illustrative conditions for a radial source, trunk and two branches. Network flows are prescribed, not hydraulically solved. The local C++ implementation has passed syntax inspection, but its original SDK runtime execution and CTest are pending an ABI-compatible shared library.

14 November 2026 | paceval Mathematical Research Demonstrator

# 1. Industrial use case and current state of the art

District heating delivers hot water from a central production plant to buildings through insulated underground supply and return pipework. Heat utility engineers need to evaluate how water loses temperature in transit, which of several consumers may receive insufficient supply temperature, and how network pipe losses and pressure requirements change when weather or operating parameters change.

Consider a hypothetical European urban district. One heat plant supplies a shared main pipe, which divides into two branches serving two customer groups. At a given weather condition the operator can examine alternative supply temperatures and estimate useful customer heat, heat lost along the pipes, and the required hydraulic head. These functions can inform planning and monitoring, but safe dispatch decisions still depend on measured demands, calibrated hydraulic network models, transient behavior and plant limitations.

## What established tools do

Network simulation tools such as DHNx and pandapipes model steady thermal and hydraulic relationships with validated, application-dependent assumptions. More elaborate approaches also handle time-dependent thermal transport, system generation units, storage and optimization. The open-source DistrictHeatingSim workflow incorporates GIS, pipe models and heat-generator design. These tools are context and potential integration partners, not software that the current research claims to replace [1,5,6].

## Recent scientific developments

A 2024 systematic review compares network modelling approaches and industrial tools [1]. A 2025 quasi-dynamic study includes travel-time effects and field measurements [2]. Publications in 2026 develop transfer-function thermal transient methods and source-network-load optimization [3,4]. These developments matter because a static pipe equation does not represent delayed temperature fronts, thermal storage or dynamic customer interactions. The present experiment deliberately studies the smaller, well-defined mathematical kernel used within such workflows.

## Research question

Can the original paceval engine evaluate reusable and versioned single-pipe and three-edge network functions, including full end-to-end heat-delivery sums, pressure-dependent pump proxies, mathematical feasibility guards and sensitivities, for independent weather and dispatch scenarios with a reproducible verification trail?

# 2. Defined thermohydraulic screening model

The original WP 13 proposes heat delivery, lumped pipe loss, Darcy friction and a supply temperature margin. Here those four kernels are retained and complemented with a distributed-loss model, two consumers, combined energy accounting, prescribed-flow hydraulic path screening, model version comparison and explicit constraints.

## Pipe temperature: two versioned steady-state models

$$h = UA / (m\ cp)\ [\text{dimensionless}]$$

$$T_{out,v1} = T_{in} - (T_{in} - T_{amb})\ h / (1 + h/2)$$

$$T_{out,v2} = T_{amb} + (T_{in} - T_{amb})\ \exp(-h)$$

The first is a midpoint-lumped loss approximation. The second is the constant-property, uniformly distributed-loss solution, as used in documented district-heating simulators [2,5]. Here UA is the total thermal conductance of a pipe section in W/K, m is water mass flow in kg/s, and cp is J/(kg K). Input and output temperatures are in degrees Celsius; differences use kelvin. The v1 domain is restricted to h less than 1 in this study.

## Thermal and hydraulic expressions

$$Q_{loss} = m\ cp\ (T_{in} - T_{out})\ [W]$$

$$Q_{delivered} = m\ cp\ (T_{out} - T_{return})\ [W]$$

$$u = 4\ m / (\rho\ \pi\ D^2)\ [m/s]$$

$$\Delta p = f\ (L/D)\ \rho\ u^2 / 2\ [Pa]$$

$$P_{pump,proxy} = (m_{total}/\rho)\ [\Delta p_{trunk} + \max(\Delta p_A, \Delta p_B)] / \eta$$

The pump expression is a deliberately simplified conservative path proxy. It presumes fixed friction factors, no minor losses and prescribed branch mass flows. It is not a converged pipe-network hydraulic solution and must not be used to size an actual pump.

## Network conservation and limits

$$m_{total} = m_A + m_B$$

$$Q_{source} = m_{total}\ cp\ (T_{source} - T_{return})$$

$$Q_{total,delivered} = Q_A + Q_B$$

$$Q_{source} = Q_{total,delivered} + Q_{loss,trunk} + Q_{loss,A} + Q_{loss,B}$$

$$M_T = \min(T_{A,out}, T_{B,out}) - T_{min}$$

$$M_Q = Q_{total,delivered} - Q_{required}$$

$$\text{Guard} = (M_T \geq 0)\ \text{AND}\ (M_Q \geq 0)\ \text{AND}\ (P_{pump} \leq P_{limit})$$

The physical return-side distribution is not simulated. All consumer return temperatures are prescribed equal to 42 °C. Heat balance concerns the explicitly modelled supply-side network and customer thermal extraction. This simplified setup isolates the mathematical graph functions rather than constituting a calibrated district model.

# 3. End-to-end mathematical execution architecture

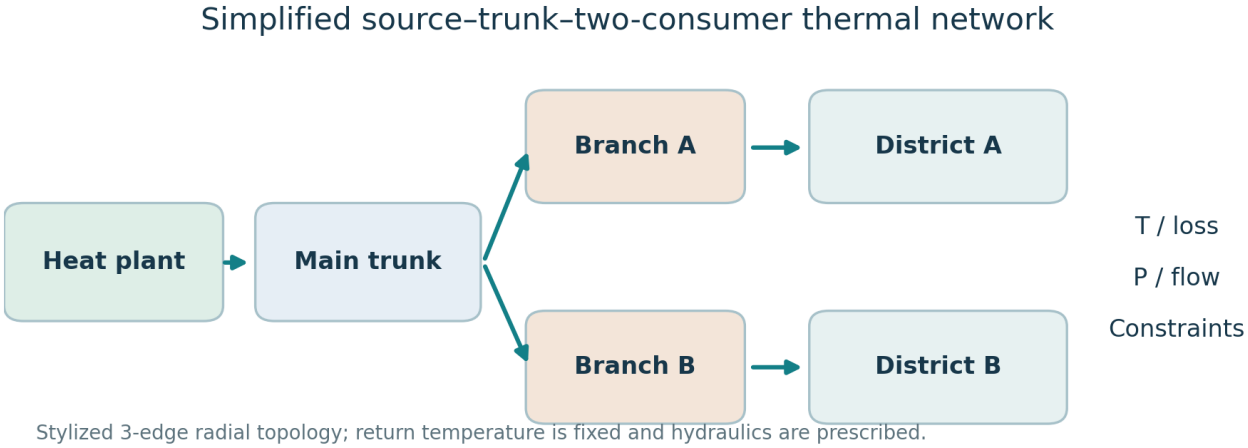



Figure 1. Conceptual architecture only. The source, trunk and two consumer branches show the algebraic topology, not a surveyed network.

The host reads validated mass flows, weather scenarios, pipe coefficients, temperature thresholds, source temperatures and load requirements. It then calls persistent paceval computation handles. The edge formulas calculate outlet temperature, loss, heat delivery, Darcy pressure drop, pump proxy and derivatives. A separate, composed network group computes the entire source-to-consumers chain directly as paceval expressions. The host decides which acceptable dispatch candidate to retain; paceval evaluates the underlying numerical scores and margins.

## Why both model versions are maintained

Model v1 uses a midpoint-lumped thermal approximation. Model v2 uses an exponential distributed-loss relation. The graph identities remain unchanged during replay; both versions are independently callable and their outputs can be compared on identical inputs. Formula versioning is explicit rather than an implicit replacement of one implementation by another.

## Meaning of operational feasibility

Feasible here means only that the mathematical inequalities for the chosen synthetic inputs are satisfied. It is not an engineering guarantee of temperature quality, pumping capability, mass-flow balance or safe operation. Boundary equality is accepted by the graph guard as stated, and the C++ host separately rejects invalid units, singular denominators and nonfinite inputs.

# 4. Original paceval integration and function coverage

The original SDK header and bridge sources are included in src/ with their original notices intact. The C++ programme creates persistent computation handles and calls paceval_dGetComputationResult, paceval_dGetComputationResultExt and paceval_dGetMultipleComputationsResultsExt. It uses the documented paceval double/string converter functions when reading and writing numeric CSV records. No compatible local shared library is present here, so these local calls are implemented and syntax-checked but not claimed as executed.

| Mathematical function / industrial use | Execution owner | Graph family or rationale |
|---|---|---|
| Two pipe outlet temperature versions | paceval | edge: outlet_v1 / outlet_v2 |
| Pipe loss, heat delivery and velocity | paceval | edge: heat_loss_*, heat_delivery_*, velocity |
| Pressure drop, pump power and margins | paceval | edge: pressure_drop, pipe_pump_power, margins |
| Analytical thermal sensitivities | paceval | edge: sensitivity_outlet_*_v2 |
| Source-trunk-branches temperature chain | paceval | network: trunk_out_* and branch*_* |
| Network total delivery, losses and conservation | paceval | network: delivered_total_*, loss_total_*, heat_balance_* |
| Combined dispatch scores and logical guards | paceval | network: guard_*, dispatch_score_v2 |
| Scenario selection, validation, CSV state | C++ host | data and state orchestration; no hidden production math |
| High precision reference and invariants | independent Python/C++ | verification only; not publication data |

The full machine-readable inventory, with formula strings, units, versions and execution owners, is provided as results/math_execution_inventory.csv. The original bridge APIs are not to be confused with the independent remote MCP Create/Evaluate/Batch service. The live service reported version 4.26 for this research.

# 5. Deterministic workload and verified result scope

The selected experimental scale is intentionally interpretable rather than a claim about throughput. The pipe sweep comprises seven ambient-temperature values and seven pipe conductances with fixed mass flow and material properties. The network sweep spans five ambient conditions and five source supply temperatures. Mass flows and heat demands vary deterministically with the weather setting, and two consumer branches are connected downstream of one trunk.

| Executed remote study | Observed and archived evidence |
|---|---|
| Unique productive mathematical graphs | 22 edge + 25 composed network = 47 |
| Pipe parameter combinations | 49, yielding 1,078 successful graph-point evaluations |
| Network dispatch combinations | 25, yielding 625 successful graph-point evaluations |
| Separate narrow model-version boundary case | 6 successful graph-point evaluations |
| TINC pointwise sub-study | 2 interval-enabled graphs, 3 cases each |
| Independent high-precision checks | 547 archived value checks passed |
| Returned pointwise TINC inclusions | 6 of 6 selected decimal records enclosed references |
| Local C++ and CTest | Syntax checked; full runtime and CTest not executed |

## Interpreting the 25 network cases

For the stipulated minimum consumer supply temperature of 68 °C, 15 of 25 synthetic network candidates satisfy the heat-delivery, minimum-supply-temperature and pump-proxy constraints, and 10 do not. The five ambient conditions are −8, −3, 2, 7 and 12 °C; the candidate source temperatures are 70, 74, 78, 82 and 86 °C. All 25 cases were evaluated in each thermal model version.

These temperatures, loads, conductances and limits are invented research assumptions, not public measurements or advice for a specific utility. The nine plotted result panels are based on stored service values; where two scenarios have identical prescribed hydraulic conditions, repeated pump values are a model property and not a measurement of operational stability.

# 6. Pipe losses and outlet-temperature maps

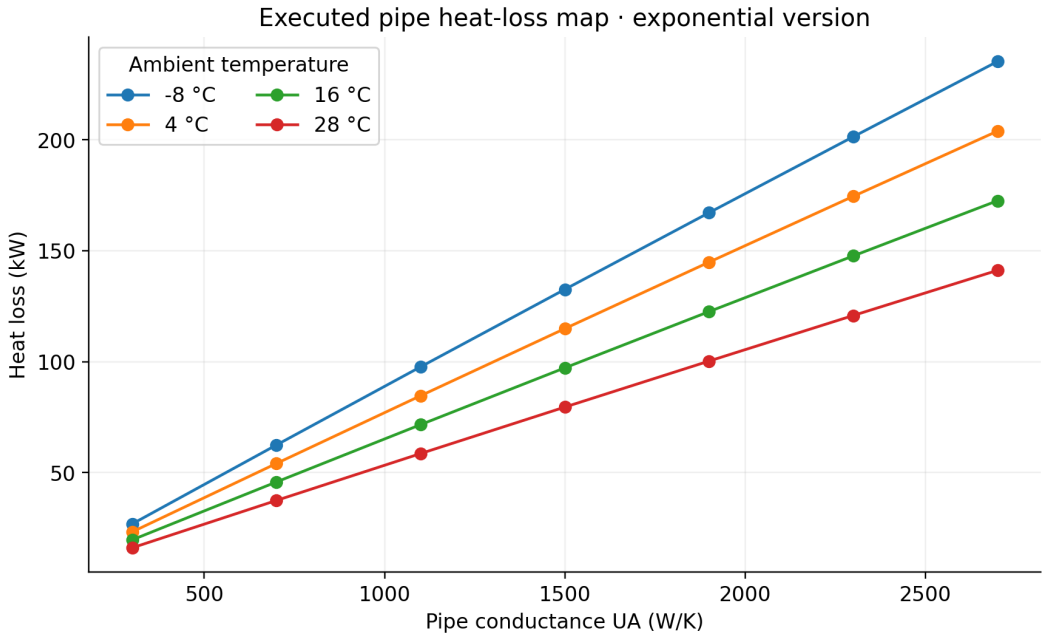



Figure 2. Recorded paceval v2 pipe losses under varying ambient temperature and thermal conductance.

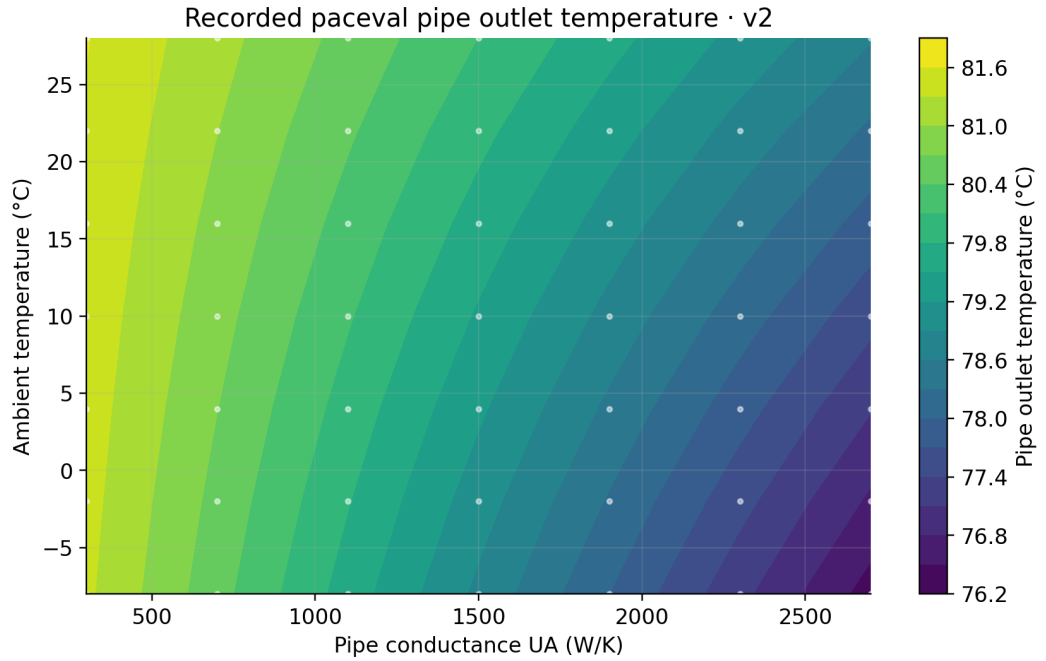



Figure 3. Pipe outlet temperatures from the actually executed 49-point paceval parameter grid.

# 7. Thermal model versions and network margins

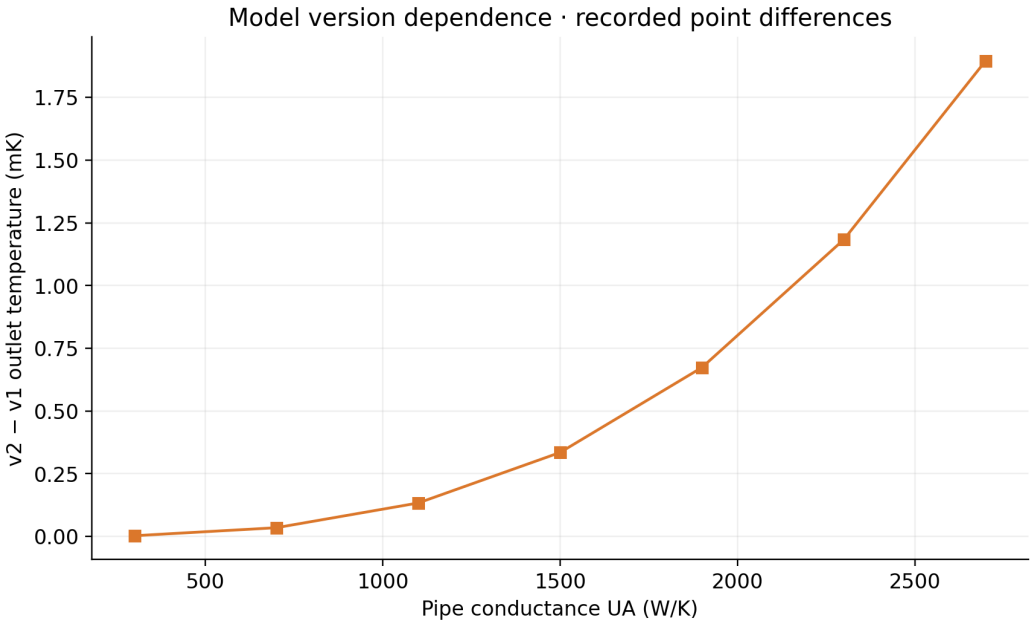



Figure 4. Difference between thermal model versions at identical inputs. This is a model-form comparison, not a numerical engine accuracy comparison.

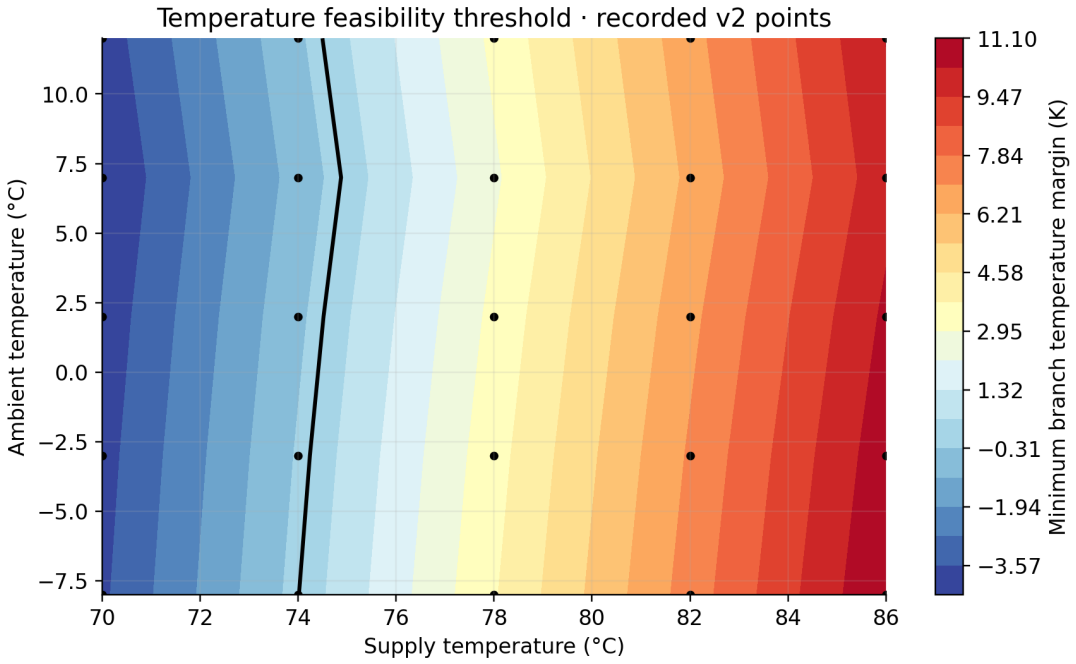



Figure 5. Minimum consumer-branch supply-temperature margin from the actual v2 network results.

# 8. Dispatch feasibility and customer thermal demand

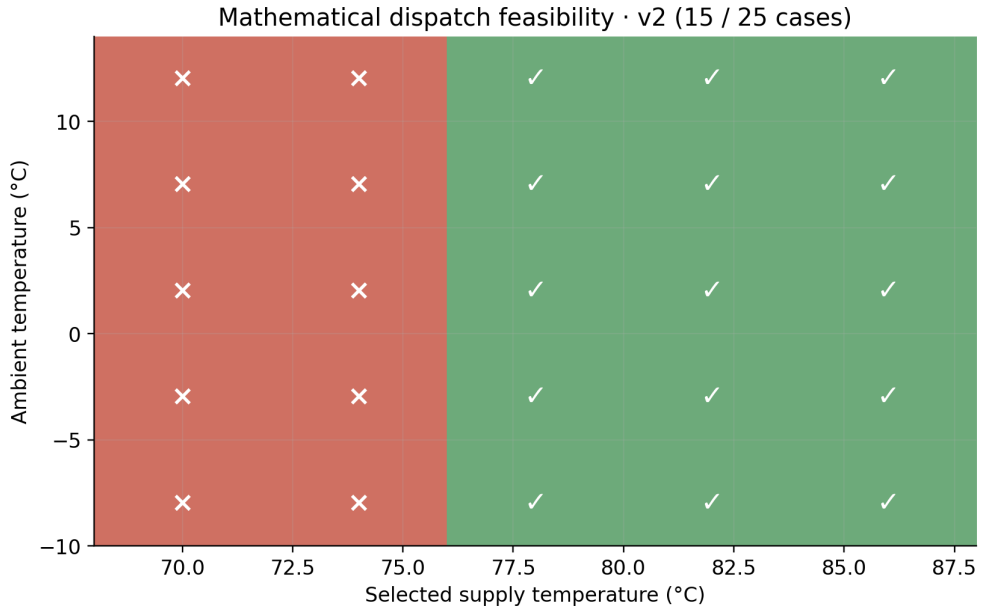



Figure 6. Mathematical acceptance of 15 network cases under the stipulated minimum temperature, heat demand and pump-proxy constraints.

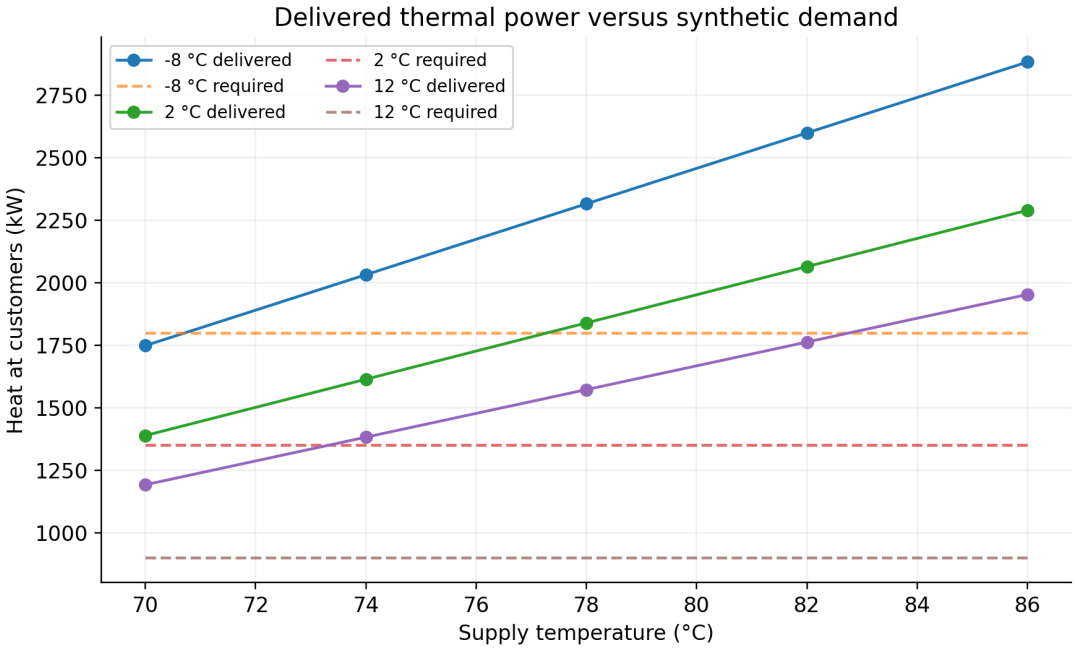



Figure 7. Delivered customer heat and synthetic demand for three representative weather scenarios.

# 9. Heat-loss fraction and hydraulic path proxy

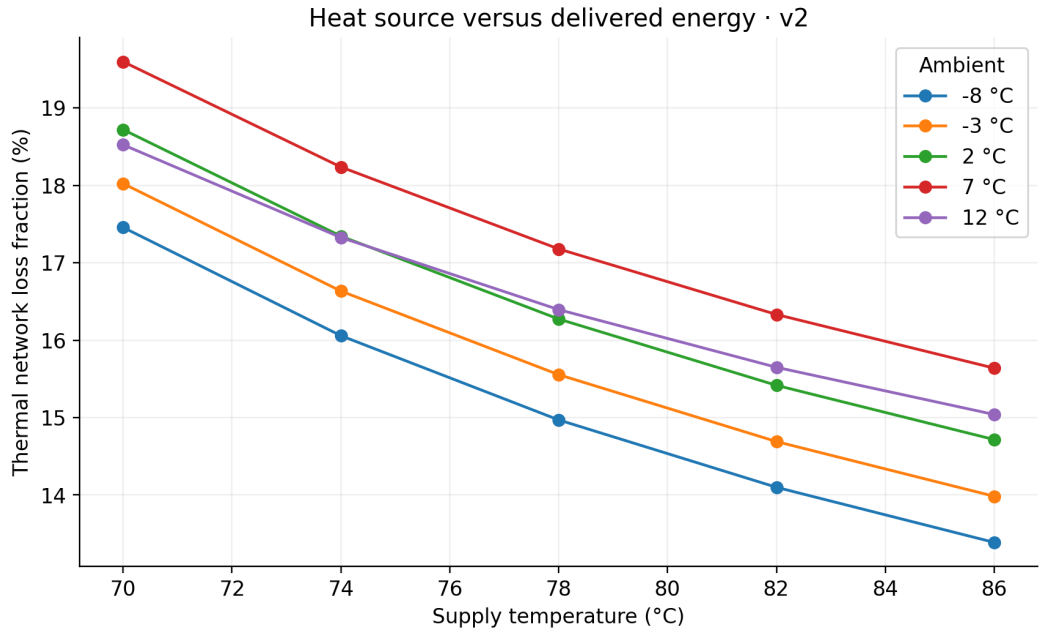



Figure 8. Ratios of source heat to modelled total thermal pipe losses across synthetic source-temperature choices.

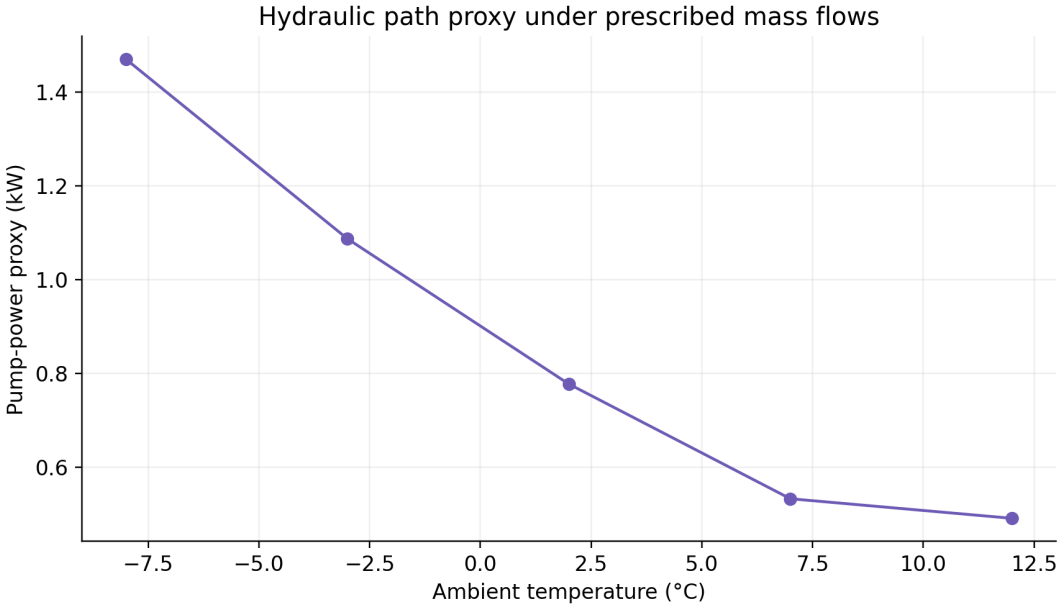



Figure 9. Prescribed-flow pump-power proxy. No dynamic flow or pressure controller is represented.

## 10. Three-dimensional dispatch parameter surface

Synthetic dispatch margin surface · actual paceval values

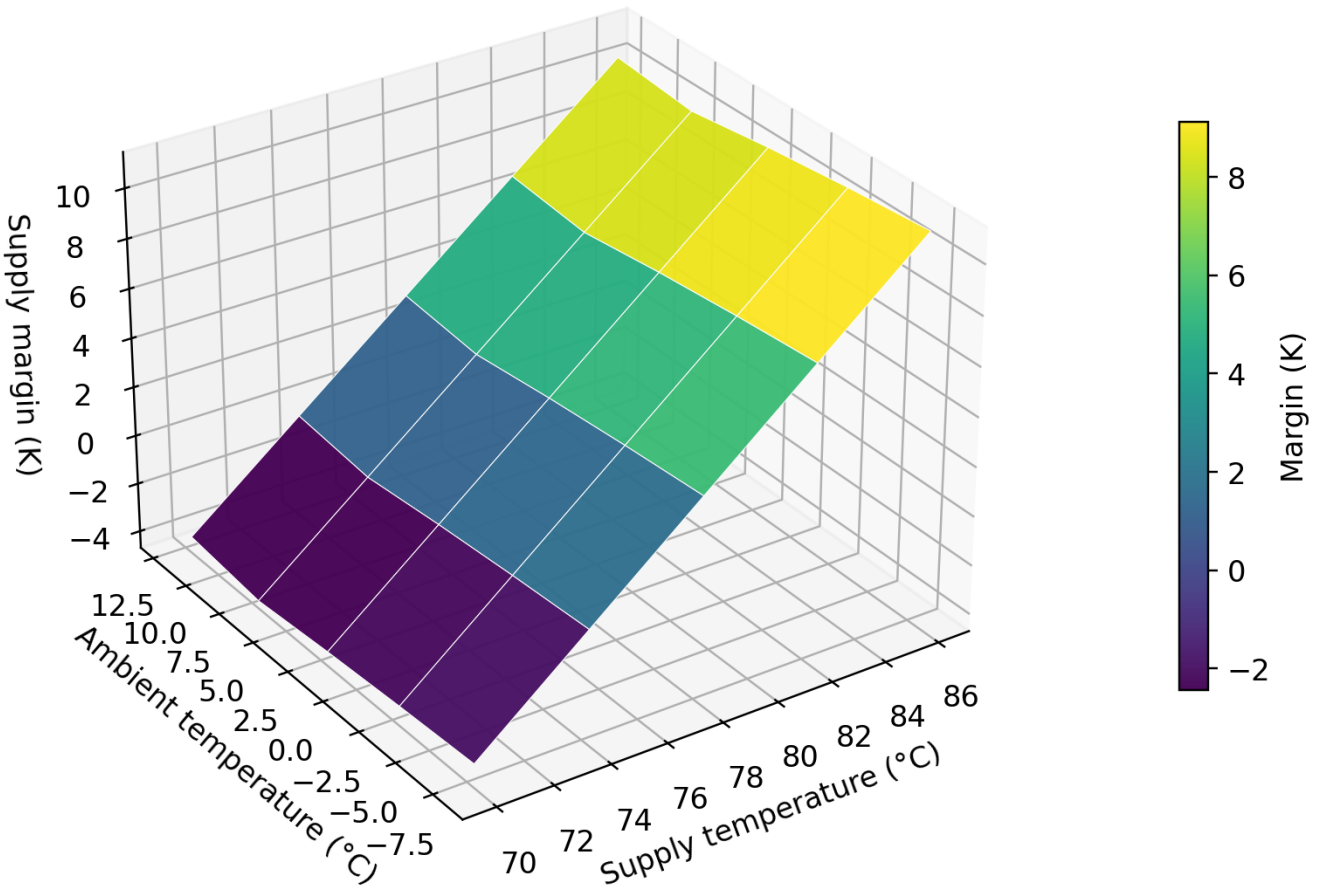



Figure 10. Three-dimensional surface joining 25 actually executed paceval temperature margins. The surface geometry between recorded points is only a plotting interpolation.

The plotted supply margin increases with source temperature for fixed weather and hydraulic inputs, while pipe heat loss depends on ambient temperature, transport conductance and mass flow. This result follows from the stipulated static heat model. It cannot establish safe operation during delayed temperature changes or transient disturbances.

# 11. Mathematical model version effect and boundary behavior

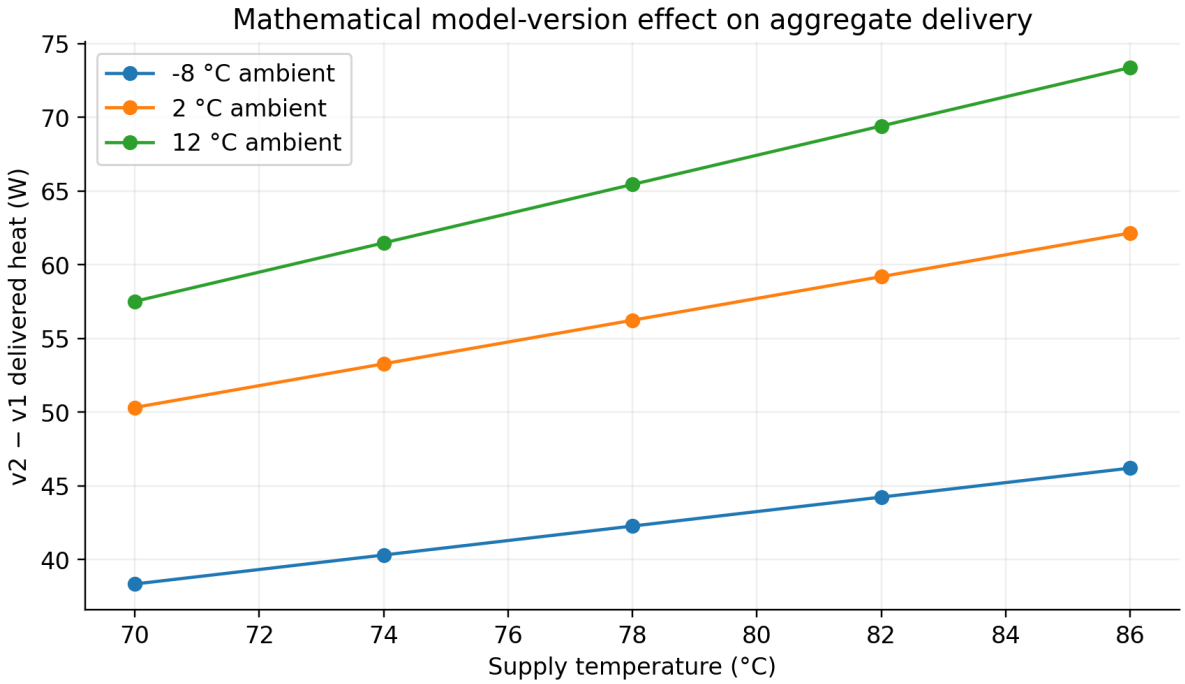



Figure 11. Recorded difference between two physical model versions for total delivered heat. Both versions were evaluated by the original paceval service.

## A deliberately narrow threshold discriminates versions

At an extra synthetic boundary case with ambient temperature −3 °C and source temperature 74 °C, the test threshold was set to 67.777 °C. The paceval midpoint-lumped version evaluated a temperature margin of approximately −0.00044016 K and a false guard, while the distributed-loss version evaluated approximately +0.00063632 K and a true guard. This precision-stress threshold is not an engineering temperature policy.

Such a flip can occur because different thermal approximations produce different physical model outputs, even when each is numerically evaluated correctly. Any real decision threshold should include measured uncertainty, model discrepancy, transient effects, calibrated demand and explicit operating policies. This example is not a claim that one model is universally correct.

# 12. Independent verification and interval boundaries

An independent 90-decimal-digit implementation reconstructed physical pipe temperatures, losses, pressure proxies, branch heat delivery, model-version margins and conservation constraints from the immutable synthetic input CSV files. It does not interpret the paceval expression strings as a second execution engine. All 547 selected archived values passed the predeclared consistency rule. Detailed numerical error data remain in results/independent_numerical_audit.json instead of being printed as an engine-accuracy comparison.

## TINC semantics

Two additional, TINC-enabled paceval expressions were executed for pipe outlet temperature and heat loss, for three fixed input states. In all six selected cases the transmitted decimal lower/upper interval endpoints enclosed the independent pointwise high-precision result. The test covers only those particular input points and text-represented interval endpoints. It does not prove that a whole range of unknown network parameters is enclosed, and does not include metrological sensor uncertainty.

## Input safety and library status

The C++ implementation uses the original documented SDK double-to-string and string-to-double interfaces. Because the numerical string parser can evaluate mathematical expressions, CSV inputs are checked against a strict decimal-literal grammar before SDK conversion. Console and CSV output pass through the SDK converter only in the prepared local execution path. The C++ code and the original SDK bridge compile under a C++17 syntax-only check. No claim is made of successful local linking, CTest, runtime converters or a published GitHub render preview.

## Limitations and next experimental validation

The model assumes steady flow, fixed pipe U and friction factors, constant cp and density, prescribed demand and branch flows, no heat losses on return pipes, no minor hydraulic losses, and no thermal propagation delay. Tests should next include transient transport, measured consumer temperatures, calibrated pipe parameters, flow reversals, zero flow, heat source limits, actual network GIS topology and comparison to full thermo-hydraulic solvers.

# 13. Conclusions and scholarly references

The original paceval Mathematical Engine demonstrably evaluated mathematical formulas for pipe heat losses, temperature propagation, pressure proxies, delivered heat, thermal model alternatives and composed dispatch margins. These functions were reused over actual recorded synthetic parameter sets, and the selected results passed an independent numerical audit. The outcome is a meaningful mathematical research demonstrator that can be studied alongside purpose-built heat-network simulation tools. It is not an industrially qualified network controller, plant optimizer, or dynamic model.

## Reproduction paths

For the recorded data, run **python validation/high_precision_reference.py** and **python scripts/make_figures.py**. To exercise the local original SDK when its compatible Linux library is available, run **bash scripts/build_and_test.sh /absolute/path/libpaceval_linux_sharedLIB.so**. A local run writes separate files under results/local_sdk_*.csv and does not overwrite the remote evidence. No shared library is distributed.

## References


[1] Design and simulation of district heating networks: A review of modeling approaches and tools. Energy 305 (2024), 132189. DOI: 10.1016/j.energy.2024.132189.

[2] A quasi-dynamic model and comprehensive simulation study of district heating networks considering temperature delay. Energy 318 (2025), 134855. DOI: 10.1016/j.energy.2025.134855.

[3] A transfer function method for thermal transient modeling in district heating networks. Energy 351 (2026), 140834. DOI: 10.1016/j.energy.2026.140834.

[4] Integrated Source-Network-Load Modeling and Optimization for District Heating Systems: Recent Advances and Future Directions. Energy Research (2026), DOI: 10.1155/er/8877364.

[5] DHNx, simulation models and steady-state pipe heat-loss equations, online documentation.

[6] DistrictHeatingSim 2.0.0, thermo-hydraulic simulation and heat-generator modelling, online documentation.

[7] HeatNetSim, open-source heating/cooling network modelling, Energy research publication.

[8] paceval Mathematical Engine, original SDK API reference and product brief.


Reference access: 14 November 2026. All new numeric research claims are linked to the supplied paceval MCP execution manifest and observed-result CSVs.